

Experiment 11: Playfair Keyspace Analysis:

```
key = input("Enter key: ").upper().replace("J", "I")
pt = input("Enter plaintext: ").upper().replace("J", "I")

alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
table = ""

for ch in key + alphabet:
    if ch.isalpha() and ch not in table:
        table += ch

print("\n5x5 Playfair Table:")
for i in range(0, 25, 5):
    print(*table[i:i+5])

# Simple demo encryption
ct = ""
for ch in pt:
    if ch in table:
        pos = table.index(ch)
        row, col = pos // 5, pos % 5
        ct += table[row * 5 + (col + 1) % 5]

print("\nCiphertext:", ct)
```

Output:

```
== RESTART: C:/Users/student/AppData/Local/Programs/Python/Python38-64/Python.exe ==
Enter key: network
Enter plaintext: hello

5x5 Playfair Table:
N E T W O
R K A B C
D F G H I
L M P Q S
U V X Y Z

Ciphertext: ITMMN
```